

0% -&gt; ADVANCED

# Java Concepts Roadmap



A visual learning journey that turns the full Java topic list into clear phases, module cards, and a simple teaching path.

**23**

Modules

**90-95%**

Java SE coverage

**5-6**

Months

```
public class Journey {  
    public static void main(...) {  
        learn();  
        practice();  
        grow();  
    }  
}
```

3 HOURS / DAY

ONE MODULE AT A TIME

REVISION + Q&amp;A

# See the whole path before you start

Eight visual phases group the 23 modules into a roadmap students can scan quickly.

01

**FOUNDATION**

Modules 1-3

02

**CORE BUILDING**

Modules 4-6

03

**OOP THINKING**

Modules 7-9

04

**EVERYDAY JAVA**

Modules 10-12

05

**UNDER THE HOOD**

Modules 13-15

06

**STRONGER JAVA**

Modules 16-18

07

**MODERN JAVA**

Modules 19-21

08

**ADVANCED VIEW**

Modules 22-23

**Start small. Build steadily.**

Focus on the current phase, then move to the next.

**24**

Teaching steps

**3h**

Daily pace

**~6 mo**

Comfortable pace

# Learn how Java thinks

Modules 1-3: introduction, fundamentals, and control statements.

## 01

## MODULE 1

## Introduction to Java

- What is Programming?
- What is Java?
- History of Java
- Features of Java
- Applications of Java
- Java Editions (SE, EE, ME)
- Java Architecture
- JDK, JRE, JVM
- Bytecode
- Java Compilation Process
- Installing Java
- Writing the First Java Program
- Structure of a Java Program

## 02

## MODULE 2

## Java Fundamentals

- Tokens
- Keywords
- Identifiers
- Variables
- Data Types
- Literals
- Operators
- Expressions
- Type Casting
- Comments
- User Input (Scanner)
- Output Methods

## 03

## MODULE 3

## Control Statements

### DECISION MAKING

- if
- if-else
- else-if
- Nested if
- switch
- Ternary Operator

### LOOPING

- for
- while
- do-while
- Enhanced for Loop
- break
- continue
- return

Programming



Java source



Compile



Bytecode



JVM

# From statements to real programs

Modules 4-6: methods, arrays, classes, objects, constructors, and core keywords.

04

## MODULE 4 Methods

- What is a Method?
- Method Declaration
- Method Definition
- Method Calling
- Parameters
- Arguments
- Return Type
- Call by Value
- Scope of Variables
- Method Overloading
- Recursion (Introduction)

05

## MODULE 5 Arrays

- Introduction
- One-Dimensional Array
- Two-Dimensional Array
- Jagged Array
- Array Initialization
- Array Traversal
- Anonymous Arrays
- Arrays Class
- Command Line Arguments

06

## MODULE 6 Object-Oriented Programming (Core)

- Class
- Object
- Object Creation
- Instance Variables
- Local Variables
- Reference Variables
- Constructors
- Constructor Overloading
- this Keyword
- static Keyword
- final Keyword

METHODS

ARRAYS

CLASSES

OBJECTS

# Think in objects, not just lines of code

Modules 7-9: OOP principles, packages, and access modifiers.

01 Encapsulation 02 Inheritance 03 Polymorphism 04 Abstraction

07

## MODULE 7

## OOP Principles

**ENCAPSULATION**

- Getters
- Setters
- Advantages

**INHERITANCE**

- Types of Inheritance
- super Keyword
- Constructor Chaining

**POLYMORPHISM**

- Compile-Time Polymorphism
- Runtime Polymorphism
- Method Overloading
- Method Overriding
- Dynamic Method Dispatch

**ABSTRACTION**

- Abstract Class
- Abstract Methods

**INTERFACE**

- Interface
- Multiple Inheritance
- Default Methods
- Static Methods
- Functional Interfaces

08

## MODULE 8

## Packages

- Package
- Built-in Packages
- User-defined Packages
- Import Statement
- Accessing Packages

09

## MODULE 9

## Access Modifiers

- public
- private
- protected
- default
- Visibility Rules

# Work with values, text, and errors

Modules 10-12: wrapper classes, strings, and exception handling.

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MODULE 10

## Wrapper Classes

- Primitive vs Wrapper
- Boxing
- Unboxing
- Auto Boxing
- Auto Unboxing

11

MODULE 11

## Strings

- String
- String Pool
- Immutable Objects
- String Methods
- StringBuffer
- StringBuilder
- String vs StringBuffer vs StringBuilder

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MODULE 12

## Exception Handling

- Exception
- Error
- Checked Exception
- Unchecked Exception
- try
- catch
- finally
- throw
- throws
- Custom Exception
- Exception Propagation

try

-&gt;

catch

-&gt;

finally

throw / throws / custom exception

# Memory, nested types, and collections

Modules 13-15: memory management, inner classes, and the Java Collections Framework.

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MODULE 13

## Java Memory Management

- Stack Memory
- Heap Memory
- Method Area
- Object Lifecycle
- Garbage Collection
- finalize()
- Garbage Collector Types (Overview)

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MODULE 14

## Inner Classes

- Member Inner Class
- Static Nested Class
- Local Inner Class
- Anonymous Inner Class

15

MODULE 15

## Java Collections Framework

### COLLECTION FRAMEWORK OVERVIEW

- Collection Hierarchy
- Iterable

### LIST

- ArrayList
- LinkedList
- Vector
- Stack

### QUEUE

- Queue
- PriorityQueue
- Deque
- ArrayDeque

### SET

- HashSet
- LinkedHashSet
- TreeSet

### MAP

- HashMap
- LinkedHashMap
- TreeMap
- Hashtable

### UTILITIES

- Iterator
- ListIterator
- Comparable
- Comparator
- Collections Class

COLLECTIONS

List

Queue

Set

Map

Utilities

# Reusable types, files, and concurrency

Modules 16-18: generics, file handling, and multithreading.

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## MODULE 16 Generics

- Generic Classes
- Generic Methods
- Wildcards
- Type Safety
- Type Erasure (Overview)

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## MODULE 17 File Handling

- File Class
- Reading Files
- Writing Files
- Buffered Streams
- Character Streams
- Serialization
- Deserialization

18

## MODULE 18 Multithreading

- Process vs Thread
- Creating Threads
- Thread Lifecycle
- Thread Methods
- Synchronization
- Inter-thread Communication
- Deadlock
- Executor Framework (Overview)

PROCESS

THREAD A

THREAD B

THREAD C



# Modern features and runtime metadata

Modules 19-21: Java 8 features, reflection, and annotations.

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MODULE 19

## Java 8 Features

- Lambda Expressions
- Functional Interfaces
- Method References
- Stream API
- Optional
- Default Methods
- Static Interface Methods
- Date and Time API

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MODULE 20

## Reflection API

- Reflection
- Class Class
- Constructors
- Methods
- Fields
- Dynamic Loading

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MODULE 21

## Annotations

- Built-in Annotations
- Custom Annotations
- Meta Annotations

input

-&gt;

lambda

-&gt;

stream

-&gt;

result

# Understand the JVM and see what comes next

Modules 22-23: JVM internals and the advanced Java overview from the roadmap.

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MODULE 22

## JVM Internals

- JVM Architecture
- Class Loader
- Class Loading Process
- Execution Engine
- JIT Compiler
- Memory Areas
- Garbage Collection Basics

23

MODULE 23

## Advanced Java Concepts (Overview)

- JDBC
- Networking
- RMI
- Servlet (Introduction)
- JSP (Introduction)
- Maven
- Java Modules (Java 9+)

### JVM INTERNALS

Class Loader

Class Loading Process

Execution Engine

JIT Compiler

Memory Areas

Garbage Collection Basics

# 24 steps from first contact to advanced overview

Use this page as the sequential teaching path provided in the source roadmap.

01

Introduction to Java

02

Java Fundamentals

03

Control Statements

04

Methods

05

Arrays

06

Classes &amp; Objects

07

Constructors

08

this , static , final

09

OOP Principles

10

Packages &amp; Access Modifiers

11

Strings

12

Wrapper Classes

13

Exception Handling

14

Memory Management

15

Inner Classes

16

Collections Framework

17

Generics

18

File Handling

19

Multithreading

20

Java 8 Features

21

Reflection

22

Annotations

23

JVM Internals

24

Advanced Java Overview

**STUDENT CHECKPOINT** Move step by step and keep revision + Q&A in the comfortable teaching pace.



ROADMAP FINISH LINE

# Choose a pace. Keep moving.

Estimated coverage: 90-95% of Java SE. Estimated duration at 3 hours/day:

~5

Months

Intensive pace



~6

Months

Comfortable teaching pace with revision and Q&A



One module at a time is still a complete roadmap.